Nature Methods

A primer to scaffolded DNA origami

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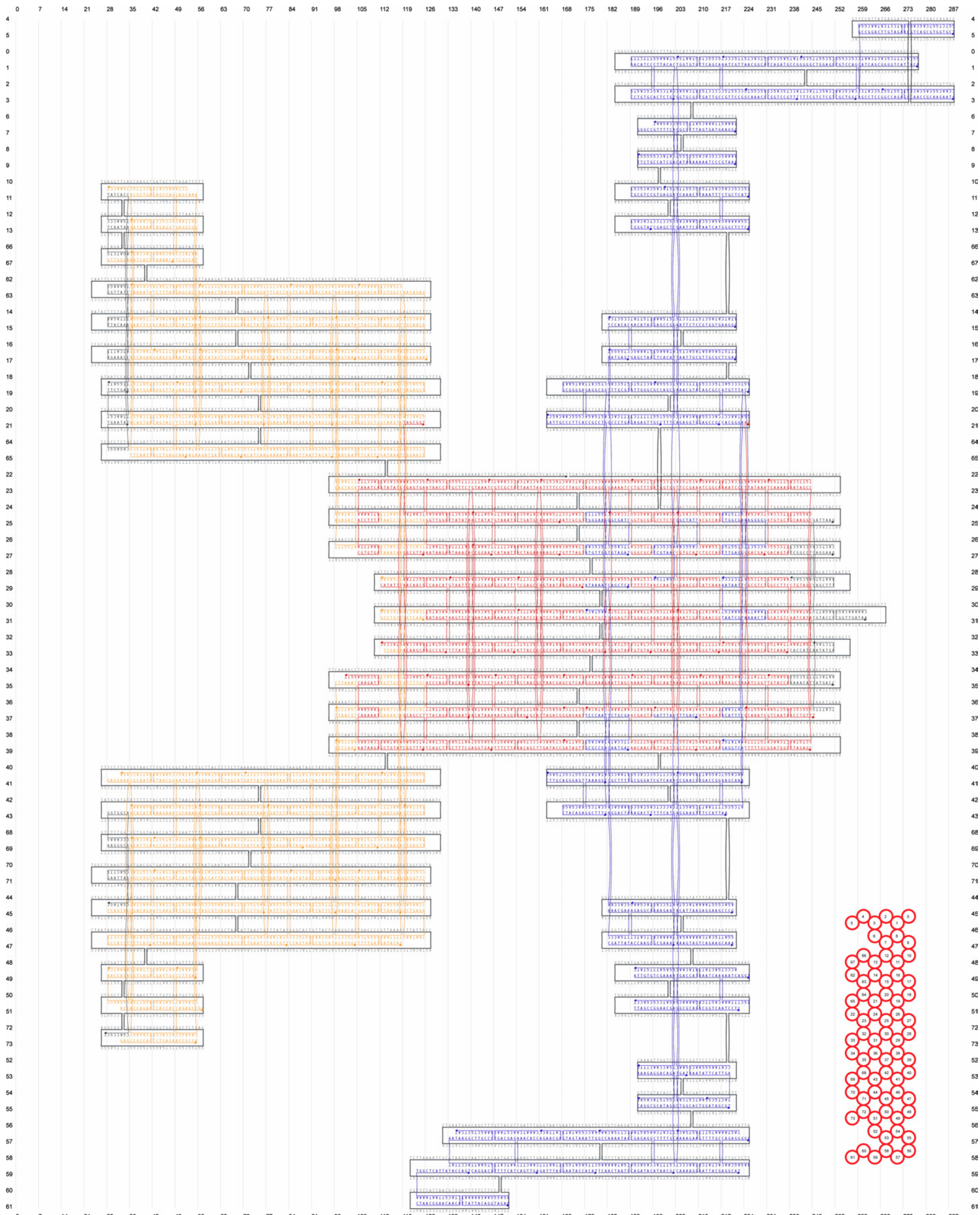
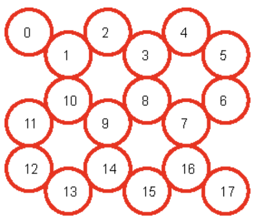
Supplementary Figure 1

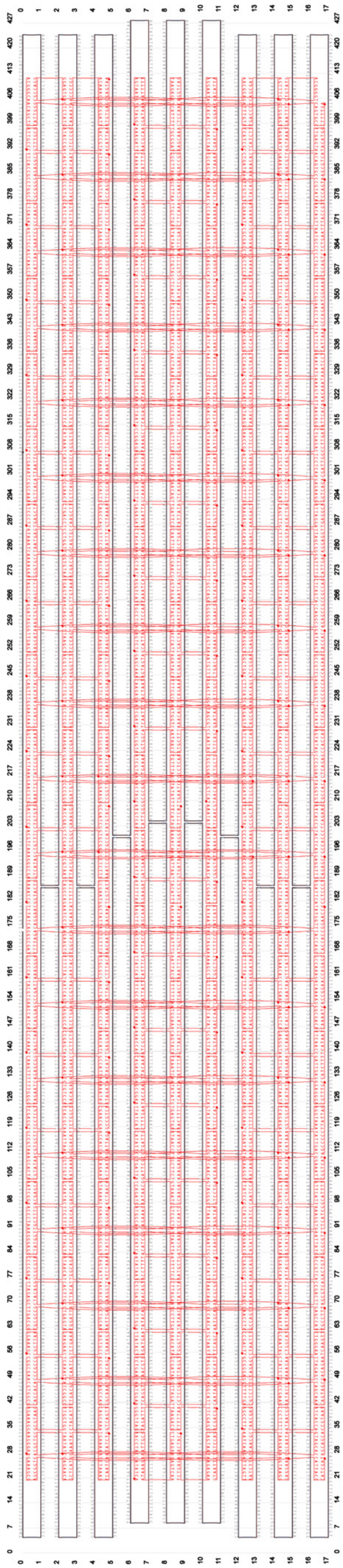


Detailed scaffold / staple lay-out for the straight ‘robot’ object. Generated with caDNAno v0.2

Supplementary Figure 2



Detailed scaffold / staple lay-out for the 18 helix bundle. Generated with caDNAno v0.2



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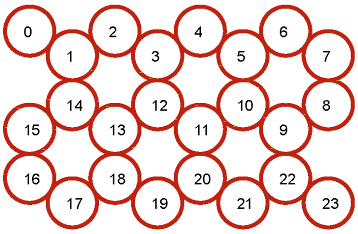
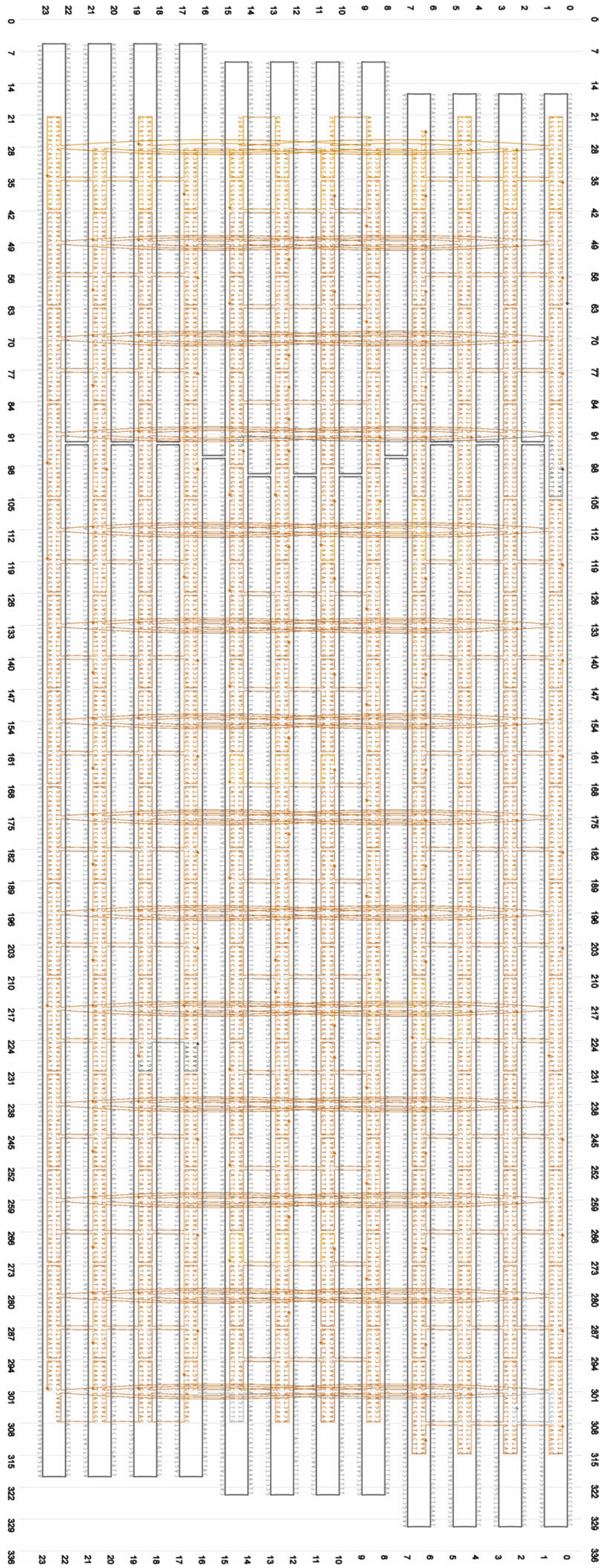
Supplementary Figure 3



Detailed scaffold / staple lay-out for the 24 helix bundle. Generated with caDNAno v0.2



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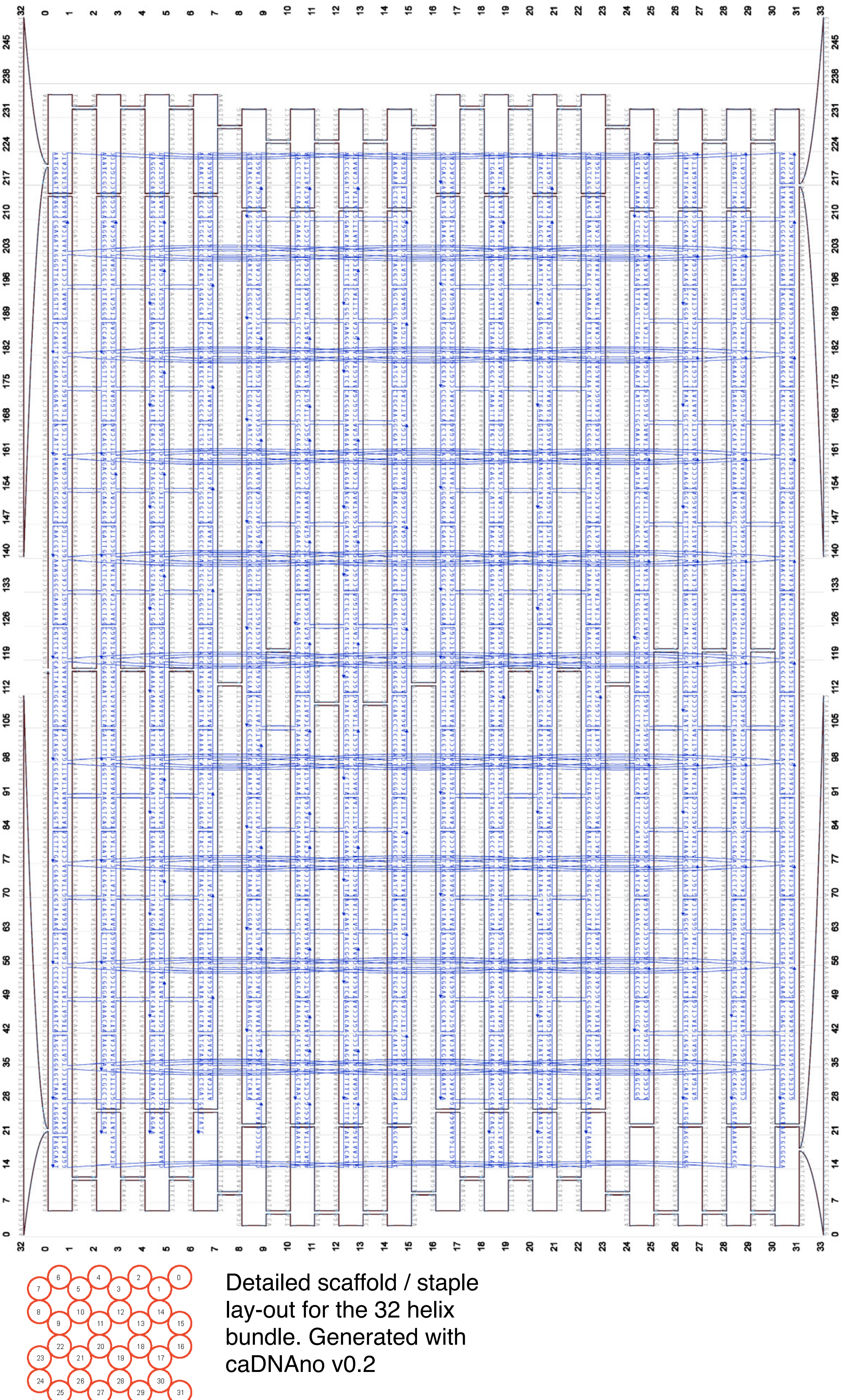
Supplementary Figure 4



Detailed scaffold / staple lay-out for the 32 helix bundle. Generated with caDNAno v0.2

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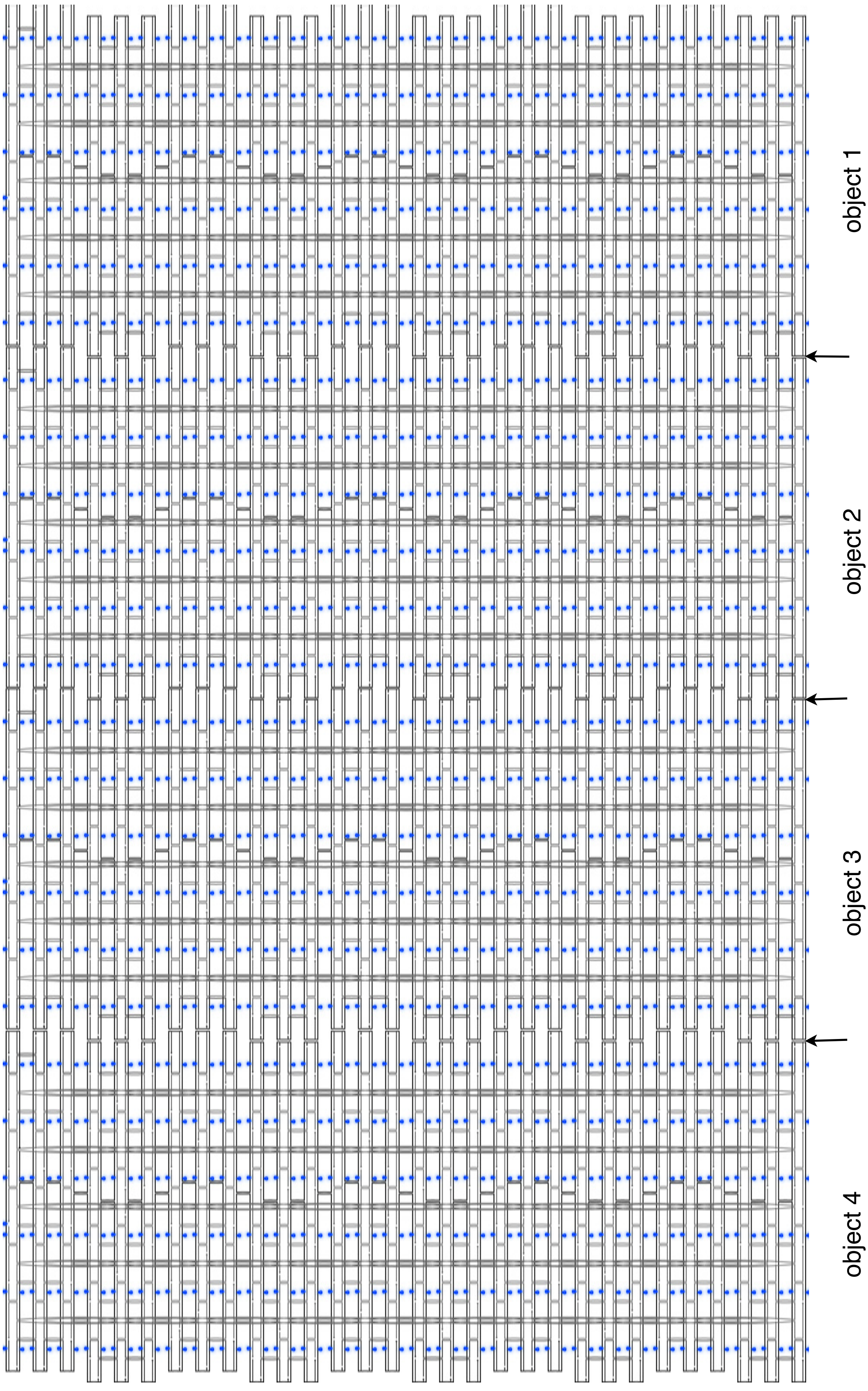
Supplementary Figure 5



Detailed scaffold / staple lay-out for a tetrameric 60 helix bundle. Original particle from Dietz H., Douglas SM., Shih WM., Science 2009. Generated with caDNAno v0.2

Nature Methods: doi.10.1038/nmeth.1570

Supplementary Protocol 1

This protocol is based on Sambrook, J., Russell, D. “Molecular Cloning: A Laboratory Manual”. Third ed. 2001, Cold Spring Harbor Laboratory Press. It was adapted and used for DNA origami purposes with JM-109 strains first in Douglas, SM., Chou, JJ., Shih, WM. “DNA-nanotube-induced alignment of membrane proteins for NMR structure determination”, *Proc Natl Acad Sci U.S.A.*, 104, 6644-6648, (2007). This is a further adaptation for the use with XL-1 Blue strains. Cite Douglas et al when using this protocol.

Step-by-step scaffold production from phage DNA.

(1) Transforming single-stranded M13-derived plasmid DNA into phage-competent E.coli cells.

- Materials:
    - Transformation-competent E.Coli strain carrying F' plasmid, or is capable of growing pili. This protocol works with XL1-Blue. JM-109 is another possible strain.
    - LB0 (Luria Bertani) medium.
    - Agarplates with tetracyclin [12,5µg/ml], IPTG [40µg/ml], and X-gal [100µg/ml]. If using JM-109 do NOT include tetracyclin in the plates.
    - single stranded phage-plasmid DNA (prepare a dilution with ~ 50 nM concentration)
    - single stranded p7249 (i.e wild-type M13mp18 DNA) as a positive control
- Procedure:
    - Obtain competent E.coli aliquots (usually 50µl, you'll need at least two if you include p7249) from -80ºC freezer, let thaw on ice for 20 min.
    - Gently pipet 5µl ssDNA on to the cell suspension.
    - Let sit on ice for 10 min.
    - Heatshock for 30 seconds at 42ºC (60 seconds for p8064)
    - Let sit on ice for another 5 min.
    - Add 200µl LB0 to the cell suspension
    - Shake-incubate at 37ºC for 30 minutes
    - Plate onto pre-warmed agarplates. Incubate agarplates overnight at 37ºC
    - Store at 4ºC.

- Expected Results:
  - On the following morning you should discover a bacterial lawn on the plates covered individual phage plaques that appear like clear "holes" in the lawn. If the phage density is very high, plaque may also cover larger areas.
  - On the wildtype p7249 plate the plaque should be blue, while on clonal variants such as p7560 or else the plaque should be clear. By the color test you can check whether inserts are still present and avoid clones that have kicked out the insert (happens, plaque turn then blue again).
- Comment:
  - XL-1 is preferred over XL-10. Plaque do also form with XL-10 but to a much lesser extent (20-fold reduction).

(2) Growing and harvesting phage in liquid culture.

Note: This protocol is written for XL-1 Blue. If using JM-109, do NOT include tetracyclin selection.

- Materials:
  - Agar-plate with M13-derived plaques
  - 10 ml LB0 containing tetracyclin in a 125 ml flask
  - 250 ml 2xYT Medium containing tetracyclin and 5mM MgCl2 (add 1.25 ml 1M MgCl2) in a 2-liter flask. Important: Use the 2-liter flask for proper oxygen circulation.
    1. Recipe for 2xYT (use flow hood):
       a. Measure ca. 900 ml of distilled water
       b. Add 16 g Bacto Tryptone
       c. Add 10 g Bacto Yeast Extract
       d. Add 5 g NaCl
       e. Adjust pH to 7.0 with 5N NaOH
       f. Adjust to 1 L distilled water
       g. Sterilize by autoclaving
  - 10 g dry PEG8000, 7.5 g NaCl
  - 10 mM TRIS pH 8.5

- Procedure:
    - Start overnight culture by scraping -80ºC XL-1 Blue aliquot with a pipet tip followed by injecting the tip directly into the 10 ml LB0-flask. Shake-incubate at 37ºC overnight.
    - Inoculate the 250 ml 2xYT flask with 3 ml bacterial suspension derived from the overnight culture.
    - Shake-incubate at 37ºC until OD 0.4 is reached. Takes about 1.5 to 2 hrs. It is important to not overgrow the culture!
    - Use the open-end of a pipet tip to scrape plaques from the agar plate prepared earlier. E.g: gently scrape a lane across an entire plate to collect reasonable amounts of phages. Inject the scraped-plaques directly into the 250-ml bacteria culture.
    - Shake-incubate at 37ºC at 250 rpm for another 4 hours.
    - Pellet bacteria by centrifugation at 4000 rct at 4ºC. Convince yourself that supernatant is clear.
    - Transfer supernatant into a fresh and clear centrifuge bottle. Add 10 g dry PEG8000 and 7.5 g NaCl.
    - Mix 30 minutes using magnetic stir bar while incubating bottle in an ice bath. If there is phage, they should precipitate at this step, causing the solution to turn cloudy.
    - Pellet phage at 4000 rcf for 15 min at 4ºC.
    - Decant supernatant, allow bottle to sit at an angle, and carefully remove remainder of supernatant using a pipette. The remaining pellet/smear should be phage particles.
    - Resuspend pellet/smear in 2.5 ml 10mM Tris pH 8.5 (or 1/100 of original volume). Active resuspension by up-and-down pipetting may be necessary.
    - Pellet residual E.coli cells by centrifugation at 15000 rcf for 15 min at 4ºC.
    - Transfer phage supernatant to fresh container. Store at -20ºC or proceed to next step.
- Expected Results:

- Step 8 should result in a cloudy solution and a visible pellet/smear should appear when pelleting the phage. If this does not happen, the yield will be probably very low. Consider re-starting from the beginning.

(3) Purify ssDNA from phage / stripping proteins by alkaline/detergent denaturation.

- Materials:
    - Lysis-Buffer (0.2 M NaOH, 1% SDS or the one from Qiagen Plasmid Prep Kit)
    - Neutralization-Buffer (3 M KOAc pH 5.5 or the one from Qiagen Plasmid Prep Kit)
    - Pure Ethanol and 75% Ethanol.
    - 10 mM TRIS-base buffer pH 8.5
- Procedure:
    - Add 2 volumes (5 ml if continuing from step (2)) lysis buffer, gently mix by inversions.
    - Add 1.5 volumes (3.75 ml if continuing from step (2)) neutralization buffer, gently mix by inversions.
    - Incubate in ice-water bath for 15 minutes.
    - Spin at 16000 rcf at 4ºC for 10 minutes.
    - Transfer supernatant into fresh container (recommend 50 ml falcon tube).
    - Add 1 volume pure ethanol (ca. 8 ml) to the transferred supernatant.
    - Incubate in ice-water bath for 30 minutes.
    - Spin at 16000 rcf for 15 minutes at 4ºC.
    - Discard supernatant. Carefully pipet-out residual supernatant.
    - Add 1.5 ml 75% ethanol to the pellet. Mix by swirling.
    - Incubate in ice-water bath for 10 minutes.
    - Spin at 16000 rcf for 15 minutes at 4ºC.
    - Discard supernatant. Carefully pipet-out residual supernatant.
    - Resuspend pellet in 1 ml 10 mM TRIS pH 8.5.
    - Use spectrophotometer to estimate concentration of resuspended ssDNA.
    - Run a 2% agarose gel containing 11 mM MgCl2 and 0.5 mg/ml EtBr. Running buffer 0.5x TBE with 11mM MgCls. Include a positive control!

- Store purified DNA at -20ºC.

- Expected Results: expect a yield of approx 1 mg ssDNA for this prep-scale at a concentration of about 350 nM.

Supplementary Protocol 2

This protocol is adapted from a recipe published in Douglas, SM., Dietz, H., Liedl, T., Högberg, B., Graf, F., Shih, WM. "Self-assembly of DNA into nanoscale three-dimensional shapes.", *Nature*, 459, 414-418, (2009).

Setting up folding reactions.

There are many different ways to set-up a molecular self-assembly reaction for scaffolded DNA origami. Many conditions have not been tried so far. Yet, there are procedures that robustly give good results when folding 3D DNA origami objects. These conditions are explained here. We seek for following standard procedures that are designed to leave little room for pipetting errors and that facilitate the comparison between results obtained for different objects or from different experimenters. Here is a short glossary to explain the terminology:

- A **folding reaction** is a mixture containing scaffold DNA, staple DNA, water, a buffer mastermix to stabilize pH, and additional salt ions (typical magnesium). **Folding** itself then refers to subjecting the sample to a thermal denaturation and renaturation procedure. We prepare folding reactions in PCR strip tubes with standardized volumes of either 50µl or 100µl, with the latter being more suited for longer thermal ramps (the amount of water loss by evaporation due to extended heating is roughly the same for 50µl or 100µl).
- A **pre-stock** is a pool of staple oligonucleotides obtained from pipetting subsets of the staples that make up a DNA origami object. A pre-stock for example may contain hundreds of oligos that form the core of an object, another pre-stock may contain only the few oligos that form the edge of an object. Pre-stock do not have standard concentrations but have a known composition and are used to set-up working stocks.
- A **working stock** is a pool of staple oligonucleotides with each staple molecule at a standard concentration of 500 nM. Working stocks are used to set up folding reactions.
- A **magnesium screen** refers to a series of folding reactions with different magnesium concentrations in the reaction mixture. Magnesium has been observed to have a drastic effect on the quality of folding of a DNA origami object and it is advisable to search for optimal magnesium concentrations. Thus, whenever a new object is being made, we run a standardized magnesium screen covering 8 different concentrations from 12 mM to 26 mM MgCl2. Optimal folding is typically observed for concentrations around 20 mM.

Step 1: Write down a pipetting guide

A useful starting point is to develop a pipetting guide for the DNA origami shape as exemplified by the scheme shown below that was prepared for a set of staple oligonucleotides on two 96-well plates named HD_P013 and HD_P014 at normalized concentration of 100 µM each. The guide is useful for preparing the pre- and working stocks. It is advisable to prepare the pipetting guide together with the staple order and to store it for later reference.

Pipetting guide for plates

HD-P013 to HD-P014 (Bioneer)

- motor_stage_v1.6

motor_stage_v1.6

PLATE	WELLS	SHAPE	MODULE	# OLIGOS	Σ
HD_P013	A1 to H12	motor_stage_v1.6	CORE	96	148
HD_P014	A1 to E4	motor_stage_v1.6	CORE	52	
HD_P014	F1 to G7	motor_stage_v1.6	EDGE	19	19
HD_P014	H1 to H2	motor_stage_v1.6	CAP RECON	2	2
HD_P014	H3 to H4	motor_stage_v1.6	GAL4 RECON	2	2
HD_P014	H5 to H6	motor_stage_v1.6	HSCRE24 RECON	2	2

working stock	composition	+ dH20 [µl] to norm to 500nM
with edge	148 µl CORE + 19 µl EDGES	33
no edge	148 µl CORE	52

scaffold is p7308

Step 3: Pipet the pre- and working stocks according to the pipetting guide

For this step one will need a multi-channel pipette and some patience. Pool for example 10µl from source wells belonging to a certain structural module of the target structure into a common reservoir. This will become a prestock. To prepare the working stock, combine stoichiometric volumes from each prestock (see the column indicated with Σ in the scheme above) into a new tube. Add water to normalize to a desired target concentration.

Step 2: Prepare folding reactions.

Decide on whether the total volume of a reaction will be 50 µl or 100 µl and whether to use 10 nM or 20 nM effective scaffold concentration in the folding reaction. By default the reactions are set up with a 10:1 excess of each staple over scaffold. Pipet 8 folding reactions in a 8x PCR tube strip and make use of a 10x MgCl2 mastermix screen strip for convenience. See detailed recipes below. Subject after thorough mixing to thermal or chemical annealing**.**

20 nM

Option 1: 100µl total, **20nM** scaffold.

Components	[µl]
Scaffold @ 100 nM	20
Staples @ 500 nM	40
10x FOBXM (50mM TRIS, 10mM EDTA)	10
ddH20	20
10x MgCl2 Screen Strip Mastermix	10
TOTAL	100

Option 2: 50µl total, **20nM** scaffold.

Components	[µl]
Scaffold @ 100 nM	10
Staples @ 500 nM	20
10x FOBXM (50mM TRIS, 10mM EDTA)	5
ddH20	10
10x MgCl2 Screen Strip Mastermix	5
TOTAL	50

10 nM

Option 3: 100µl total, **10nM** scaffold.

Components	[µl]
Scaffold @ 100 nM	10
Staples @ 500 nM	20
10x FOBXM (50mM TRIS, 10mM EDTA)	10
ddH20	50
10x MgCl2 Screen Strip Mastermix	10
TOTAL	100

Option 4: 50µl total, **10nM** scaffold.

Components	[µl]
Scaffold @ 100 nM	5
Staples @ 500 nM	10
10x FOBXM (50mM TRIS, 10mM EDTA)	5
ddH20	25
10x MgCl2 Screen Strip Mastermix	5
TOTAL	50

Components	[µl]	[µl]	[µl]	[µl]	[µl]	[µl]	[µl]	[µl]
MgCl2 @ 500 mM	24	28	32	36	40	44	48	52
ddH20	76	72	68	64	60	56	52	48
Total Volume	100	100	100	100	100	100	100	100
Effective 1x Concentration	12	14	16	18	20	22	24	26

Supplementary Protocol 3

This protocol is adapted from Douglas, SM., Dietz, H., Liedl, T., Högberg, B., Graf, F., Shih, WM. “Self-assembly of DNA into nanoscale three-dimensional shapes.”, *Nature*, 459, 414-418, (2009).

Agarose Gel Electrophoresis (EtBr) with DNA origami objects

Gel Preparation

Prepare a 2% agarose gel. (Here we assume 125 ml gel slab volume)

- weigh 2.5 g agarose ultra pure (Invitrogen) into a beaker
- fill up to 125 g with 0.5x TBE-Puffer
- boil in microwave until the agarose is completely dissolved (2 min @ 800 watts)
- fill up again to 125 g with ddH2O
- cool it under the water tap until hand warm
- add 1 ml of 1.375 M MgCl2 solution (gives total MgCl2-concentration of 11mM)
- add 7 µl ethidium bromide from a 10 mg/ml stock solution)
- fill the gel tray and install desired comb immediatly
- once gel is solid, fill the gel box with TBE/11mM MgCl2 buffer
- remove comb

Gel Loading

(applies for folding-reactions containing ~10nM scaffold DNA)

sample preparation

- mix gently 12 µl sample with 3 µl 6x-loading dye (containing at least 30% glycerole)
- pipet the mix into a gel pocket

reference preparation

- mix
  - 1.2 µl single-stranded phage DNA (@100nM)
  - 10.8 µl ddH2O
  - 3 µl 6x-loading dye
- load the mix into a gel pocket

ladder loading

- load 6 µl of the premixed ladder (2log or 1kb, New England Biolabs)

Running the Gel

- put the gel box into a ice water bath
- apply U=70V const.
- REMINDER: negative DNA segregates toward the (red) plus pole
- suitable segregation and imaging after ~3-4h

Crunch-n-Squeeze Origami Purification

Comment

- Use UV transilluminator for band visualization if gel was stained with EtBr.
- Wear goggles and long sleeves to protect yourself from harmful UV radiation.

Procedure

- Cut out the desired band with a razor blade. Remove excess agarose.
- put agarose slice into a 1.5ml tube
- crunch the agarose slice with a pistil
- spin down the agarose debris
- cut-off the debris containing tip of the tube
- put the inverted tip into a freeze'n'squeeze spin column (Biorad)
- spin 10 min at full speed in a tabletop centrifuge.

Supplementary Protocol 4

We were unable to identify the original inventor of the recipe for uranyl formate stain solution. However, a version of this protocol was first used to image DNA origami nanotubes by Douglas, SM., Chou, JJ., Shih, WM. “DNA-nanotube-induced alignment of membrane proteins for NMR structure determination”, *Proc Natl Acad Sci U.S.A.*, 104, 6644-6648, (2007). This is a revised form of the recipe that adds a convenient option for longer term storage of the uranyl formate staining solution.

Negative-Staining for Transmission Electron Microscopy

(1) Prepare 2% Uranyl-Formate stain solution with 25 mM NaOH

- Materials:
  - 0.1 g Uranyl-Formate powder. Always keep container in the dark and minimize exposure to light!
  - 10 ml ddH20
  - 5 µl 5M NaOH
  - 0.2 µm syringe filter
  - 10 ml syringe
  - 12 ml falcon tubes (2x)
  - aluminum foil
  - 1.5 ml eppendorf tubes
- Procedure:
  - Bring the ddH20 to boiling. Keep boiling for 2 to 3 minutes to de-oxygenate.
  - Weigh-out 0.1 g UFo into a 12 ml falcon tube
  - Add 5 ml of still-hot ddH20 to the UFo powder. Tightly close lid, wrap in aluminum foil and shake/vortex rigourously for 10 minutes (for instance, fasten tube to a vortexer). If the UFo powder was rather fresh, you should obtain a turbid - yellowish solution. As the UFo ages (through light exposure, presumably), the solutions typically appear more brownish.
  - Filter solution through the 0.2 µm syringe filter. It should become clear now.
  - Aliquot 1 ml of the solution into four separate eppendorf tubes.
  - Centrifuge at max speed for 5 minutes in a table top centrifuge.

- Wrap three of the four tubes in aluminum foil and freeze for later use.
- Add 5 µl of 5M NaOH to the remaining tube with 1 ml of stain solution and vortex immediately for two to three minutes.
- Spin again at top-speed for 3 minutes in a table top centrifuge.
- Wrap tube in aluminum foil, keep lid clear.
- Stain solution is now ready for immediate use!

- Comment:
    - UFo solution can be stored at room temperature for about ten days in the dark if no NaOH is added. After longer periods of storage precipitates / little crystals start to appear, rendering the stain unusable for TEM. NaOH speeds up the degradation process, thus only add NaOH for immediate use.

(2) Staining samples

- Materials:
    - Whatman filter paper no 1 or No 2.
    - tweezers / forceps
    - parafilm
    - ddH20
    - 2% UFo staining solution
    - carbon-coated TEM grids (e.g EFCF400-Cu-50, Science Services, Munich, Germany)
    - Sample solution
- Procedure:
    - Glow-discharge grids (e.g Glow discharger EMS 100, Electron Microscopy Sciences). One may have to test different exposure settings in order to make the surface hydrophilic and sticky for DNA origami objects. Let the grids cool to room temperature again.
    - Put some water droplets on the bench and stick a 10x10 cm parafilm piece to it. This is now your clean working area.
    - Grab a TEM grid with forceps.
    - Apply ~3 µl sample solution onto the carbon-coated side of the TEM grid. Let adsorb for a while (3 to 4 minutes are good for ~ 1nM dilute sample solutions).

- Meanwhile, apply a 25 μl stain solution droplet onto the para film.
- Use filter paper edge to drain excess liquid from the edge of the grid. Do not touch the grid surface. Remaining liquid film will evaporate in a few seconds, so act quickly now.
- Immerse grid sample-side first into the stain-solution droplet. Incubate for 40 seconds.
- Use filter paper edge to dap off excess liquid from the edge of the grid. Do not touch the grid surface.
- Let the grid dry completely before injecting into the TEM! (30 minutes)
- Ready for imaging!

- Comments:
  - Depending on the carbon quality, sample concentration / composition, incubation times may have to be varied. Washing the grid with 0.5 $MgCl_2$ prior to applying the sample can influence orientations on the grid (more interface views).

Preparing Custom Grids for Transmission Electron Microscopy

(1) Coat grids with a thin collodion plastic film.

- Materials:
  - Collodion
  - minimum 20 cm diameter bowl/basin
  - ddH20
  - TEM copper grids 200 or 400
  - vellum paper ('Pergament-Papier')
  - tweezers, forceps
  - razor blade
  - 200 pipet
- Procedure:
  - Completetly fill basin with ddH20 right to the rim
  - Use razor blade to shorten a 200er pipet tip to create a 3mm diameter aperture
  - Pipet out 50μl collodion solution.

- Eject the collodion from approx 10 cm onto the calm water surface. CAUTION: collodion is toxic! Ensure proper ventilation and hold breath while handling the collodion!
- A collodion film will spread across the water surface and harden. Wait for 5 min.
- Use forceps to position copper grids onto the collodion film. Put shiny (specular-reflecting) side of the grids facing the collodion. Arrange in a neat hexagonal or square pattern.
- Position vellum paper on top of grids. Let soak with water.
- Cut collodion film around the circumference of the vellum paper by using the razor blade.
- Carefully pick up the vellum paper at one corner/edge and pull out of the water.
- Turn the vellum paper with the grids facing up, put into petri dish, partially cover and let dry overnight in a clean, dust free area (i.e. in a hood)
- Ready!

(2) Evaporate carbon film onto grids.

Supplementary Protocol 5

This protocol was kindly contributed by P. Rothemund, Caltech, U.S.A and used in Rothemund, PWK. “Folding DNA to create nanoscale shapes and patterns.”, *Nature*, 440, 297-302, (2006). Please cite accordingly if used.

Imaging DNA origami shapes with atomic force microscopy

Single-layer DNA origami shapes have been successfully imaged using tapping or peak force tapping mode for example on a Multimode VIII AFM with an E-scanner (Veeco Instruments, Santa Barbara, USA) and on Asylum devices (such as the MFP-3D) equipped with DNP-S oxide sharpened silicon nitride or SNL sharp nitride (Veeco Probes, Camarillo, USA) cantilevers.

Imaging can be performed in the buffer in which the shapes are being formed (TAE/Mg^{2+}) but objects may move on the surface. To overcome this problem, one can wick off the deposition buffer and add buffer with 5-10 mM nickel acetate (or chloride) which will stick the origami very strongly to the surface and may help overcoming mechanical stability problems.

DNA Origami objects can also be imaged dry. Rinse the sample with a few drops of distilled/deionized water to get rid of salts and then blow dry with canned air or nitrogen. There will be fewer origami (nickel added post-deposition would probably help) and some coffee-stain type drying artifacts with shredded origami (in general the prep will be more irreproducible) but it should still work just fine if your local AFM person likes imaging in air better. Slightly lower resolution, on average, will be achieved.

Surface patterns as produced for example by dumbbell hairpins on single-layer DNA origami rectangles are harder to reproducibly image than structures with physical holes (e.g. Paul’s smiley face or the triangle). In buffer the hairpins move around under the influence of the tip and "disappear" if tapping is too hard (the tip just moves them out of the way). With a dry sample, or under alcohol (after drying and rewetting with alcohol) the hairpins are more reproducibly visualized because they don't move, but then resolution goes down. Gross patterns (say with columns or rows of at least three hairpins in a row) are much more easily visualized than patterns with sparse hairpins, if you are going to go this route.

Supplementary Note 1

Refer to Douglas, SM., Marblestone, AH., Teerapittayanon, S., Vazquez, A., Church, GM., Shih, WM. “Rapid prototyping of 3D DNA-origami shapes with caDNAno”, *Nucleic Acids Res*, 37, 5001-5006, (2009) and please visit http://cadnano.org for online tutorials to the design of DNA origami shapes with caDNAno. We provide here a short introduction that may help understanding the concept of caDNAno-generated scaffold / staple lay-outs and source files.

Understanding caDNAno diagrams and source files

A caDNAno DNA origami design diagram consists of numbered rows that each contain a horizontal array of two squares, one drawn on top of the other. A single row is a placeholder for a single DNA double helix running through a DNA origami shape. In a way, one may call the rows in the caDNAno diagram "virtual helices" since they give a name to a potential helix in the origami shape - yet, this helix may have interruptions or have segments that are single-stranded.

The squares in a row should be considered as containers for bases contributed by either scaffold or staples. Each of the pair of squares in a row is to indicate a position where a DNA basepair can be positioned. A DNA double helix is formed from two strands: "scaffold" strand and "staple" strands, hence two squares, one on top of the other to indicate a potential basepair. The source file specifies the connectivity of the nucleotides that may be present in these containers. By default, a container will host a single nucleotide whose 5' and 3' end will have a user specified connectivity, yet the container may contain more than just one nucleotide. The container may also contain no nucleotide.

A caDNAno diagram follows some parity conventions. These parity conventions make sense in light of DNA origami making use exclusively of antiparallel holiday junctions (strand crossovers). In order to connect two neighboring DNA double helices, the strand that makes the connection must change direction upon crossover. Hence a convention is made that states that even-numbered helices have staples running 3' to 5' (left to right in the diagram) while odd-numbered helices have staples running 5' to 3'. Correspondingly, on even numbered helices, the scaffold runs from left to right in 5' to 3' direction, while on odd numbered rows the scaffold runs from 3' to 5'. In the origami shape each DNA helix has nearest neighbors where the strands have opposite parity. Regardless of parity, on each helix (row in the diagram), the top square of the duplex-square-array always refers to a strand running from 5' to 3', while the bottom square refers to a strand running 3' to 5'. Thus, the scaffold portion of a DNA double helix is drawn in the top squares on even rows (helices), while it is drawn in the bottom squares on odd rows. The scaffold portion in a row is always labeled in cyan-blue, while the staple color may be chosen by the user.

If a square is not empty, then this indicates that this position will be actually populated by a nucleotide. \footnote{A square may contain more than just one nucleotide, indicated by a little loop with a number. The number of nucleotides in the square is thus 1+ this number. If the partner staple square is also non empty, then by convention the square with a loop reflects a double-helical stretch of length (1 + number) basepair. Insertions can be used to create internal stresses that can induce curvature or twist of the final DNA origami object.}

If both top and bottom square are not empty, then this indicates a complete DNA basepair. If only one of the pair of squares is non-empty, then this indicates a single DNA base. Each nucleotide has a physical 5' and a 3' end on its backbone. If an individual square is filled by a simple line, then this indicates that this nucleotide is physically connected to two neighboring nucleotides on both 5' and 3' end. A filled colored box in a square indicates that the nucleotide is only connected to a neighboring base on its 3' end, while a filled triangle indicates that the nucleotide is only connected on its 5' end. The triangles point towards the next possible 5' end of a neighboring nucleotide. These situations corresponds to nicks in the phosphate backbone of a DNA strand.

Occasionally (actually quite regularly), one will see lines drawn vertically that connect a square within a row to another square in a distant row. This is to indicate that the nucleotides in those squares are physically connected on one of their two physical backbone ‘outlets’ to a nucleotide sitting in a neighboring DNA double helix rather than the same one. In short, this is to indicate a strand crossover. It is important to remember that the diagram is actually a side view of the final structure laid out into two dimensions. Although the vertical arcs may be quite long in the diagram, the strand crossovers that are indicated by these arcs are simple backbone U-turns that are used to connect neighboring DNA double helices in the origami shape.

Interpreting a caDNAno source file

One can read and edit the source file using any text editor. The object "vstrands" contains the details of the design. Each helix in the diagram is specified separately within a set of { } brackets. The properties "row" and "col", for example specify the position of a helix on the honeycomb raster pattern, while "num" declares the index of a helix, i.e. in which row if is to be found in the diagram. The parity of "num" decides on the polarity of the scaffold and the staple portion of the helix as described above.

The declaration "scaf" specifies the details of each nucleotide of the scaffold strand within each helix. The bases are labelled as pointers in the format '[V_0,b_0,V_1,b_1]'. The variables 'V_0' and 'V_1' indicate the initial and final strands that a certain base connects to, and 'b_0' and 'b_1' indicate the initial and final base. As such, a statement of '[0,63,0,65]' indicates that this particular base has a connection from strand 0 to strand 0 (same strand) and from bases 63 to 65. '[0,63,1,64]' indicates that this particular base is connected to base 63 on strand 0 (same strand) and to base 64 on strand 1. Hence, we can deduce that this must be base 64 on strand 0, but we further deduce that this base is followed by a strand crossover. A readout of '-1' indicates that either the base connects to or from nowhere. For example, [-1,-1,0,1] on an even-numbered helix would indicate a scaffold base that is connected on the 3' end to base 1 on strand 0, but has no connection on its 5' end. If the array element are given by '[-1,-1,-1,-1]', then there is no base present. The array "scaf" has exactly as many elements of the type '[V_0, b_0, V_1, b_1]' as squares are drawn in the diagram. Whenever a square is non empty, the array entries will differ from [-1,-1,-1,-1]. Staples are declared similarly in the array "stap".

As described above, by default each array member in "stap" or "scaf" refers to a single base if the entry differs from [-1,-1,-1,-1]. The array "loop" declares deviations from this default convention that are used to create internal stresses in the origami shape, that ultimatively lead to global twisting or bent conformations ("loop" is somewhat of a misnomer, it should be called "extensions" or "insertions"). A strand that has no loops (i.e local insertions) will have an array following the declaration that is entirely composed of 0's. The number of extra bases will be indicated by any integer greater than or equal to 1, with the location in the array marking the index of the base where the insertion will be introduced. For example '[0,0,2,0,0,0 ...]' indicates the presence of 2 extra bases at the location of the third base (base-index 2). It is important to see this in combination with the "scaf" and "stap" declarations.

Example: Consider the following declaration

"num":0 ,

"scaf":[[-1,-1,0,1],[0,0,0,2],[0,1,0,3],[0,2,-1,-1]],

"stap":[[-1,-1,-0,1],[0,0,0,2],[0,1,0,3],[0,2,-1,-1]],

"loop":[0,0,2,0,].

This is now intepreted as follows: This is helix 0 (thus, even-numbered), hence scaffold runs from 5' to 3' and staples run 3' to 5'. The "scaf" declaration tells us that the scaffold strand starts of with an unconnected 5' end on strand 0 and base-index 0, then simply goes straight from base to base on helix 0. The scaffold ends with an unconnected 3' end on base-index 3.

Importantly, the "loop" array has a non-zero entry on base-index 2. This means, that instead of populating the third entry in the "scaf" array with just one nucleotide, this "base-position will be populated by 1 + 3 nucleotides. Inlet-and outlet of this stretch of 4 nucleotides are still as specified: the 5' end is connected to base-index 1 on strand 0, while the 3' end is connected to base-index 3 on strand 0. In this example, the "stap" array has the exactly the same entries as the "scaff" array. Hence, this staple will be fully complementary to the scaffold strand. By convention, entries in the loop array also extend the basepair-content in the "stap" array. Thus, this example statement produces a double helical stretch of length 6 basepairs on helix 0 - even though only 4 entries are made in the array.

Skips (deletions) in the strands are indicated similarly, with the declaration "skip" preceding the array, with the convention that skips never exceed the value of 1.

Example: Consider the following declaration (scaf and stap the same as above)

"num":0 ,

"scaf":[[-1,-1,0,1],[0,0,0,2],[0,1,0,3],[0,2,-1,-1]],

"stap":[[-1,-1,-0,1],[0,0,0,2],[0,1,0,3],[0,2,-1,-1]],

"skip":[0,0,-1,0,].

This statement deletes the scaffold nucleotide with base-index 2 that connects from 0,1 to 0, 3. By convention, it also deletes the staple nucleotide on base-index 2. This example produces thus a double helical stretch of length 3 basepairs. Importantly, it is perfectly identical to the following statement:

```
 "num":0 ,
 "scaf":[[-1,-1,0,1],[0,0,0,3 ],[-1,-1,-1,-1],[0,1,-1,-1]],
 "stap":[[-1,-1,-0,1],[0,0,0,3 ],[-1,-1,-1,-1],[0,1,-1,-1]],
"skip":[0,0,0,0,]
"loop":[0,0,0,0,].
```

Note the change in connectivity of the entries with index 1 and 3.

Supplementary Note 2

Please refer to http://cando.dna-origami.org for submitting files for analysis and additional resources.

Computer-aided engineering for DNA origami: 'CanDo'.

1. DNA model

DNA double helices are modeled as continuous and homogeneous, isotropic Euler–Bernouilli beams characterized by stretch, twist, and bend moduli (1). This approximation ignores potential sequence-specific mechanical properties. The DNA double-helix is assumed to have axial rise 0.34 nm/bp and diameter 2.25 nm (2) with stretch modulus 1100 pN, bend modulus 230 pNnm^2, and twist modulus 460 pNnm^2 (3-7). Twist-stretch coupling is ignored in the current implementation (3). Strand crossovers that couple deformations of neighboring helices are modeled as rigid links that fully constrain the stretch, twist, and bend degrees of freedom of joined helices. Two-node Hermitian beam finite elements are used in the finite element method, which describes stretch, bend, and twist deformations using three translational degrees of freedom, two rotational degrees of freedom for bending, and one rotational degree of freedom for twisting at each finite element node along the neutral axis of the beam.

2. Prediction of 3D structure

caDNAno design files are parsed by CanDo to read geometric data including helix location on the cubic or honeycomb lattice, locations of inserted and/or deleted basepairs, and crossover positions. An initial configuration is first generated where all helices are arranged linearly and co-axially in space. Next, finite element nodes of basepairs that are coupled by inter-helical crossovers are displaced axially to the corresponding crossover positions based on the crossover spacing rule from the average helicity of B-form DNA. In this deformed configuration, rigid links are placed between nodes joined by a crossover, after which all external loads are released so that the structure is allowed to deform to satisfy the full equations of equilibrium (8). The deformation analysis may be performed using either geometrically linear or nonlinear analysis (8).

3. Calculation of 3D structural flexibility

Root-mean-square fluctuations (RMSF) of the deformed 3D structure in thermal equilibrium are calculated in the standard way (9,10) using the equipartition theorem of statistical mechanics and normal mode analysis (NMA). The RMSF for each FE node *i* is calculated by considering contributions from 200 low frequency normal modes using $\langle \Delta \boldsymbol{r}_i^2 \rangle^{\frac{1}{2}} = \left(\sum_k \langle \Delta \boldsymbol{r}_{ik}^2 \rangle \right)^{\frac{1}{2}} = \left(\sum_k \frac{k_B T}{\lambda_k} \boldsymbol{\xi}_{ik}^2 \right)^{\frac{1}{2}}$ where $\boldsymbol{\xi}_{ik}$ is the displacement vector of node *i* due to normal mode *k*, λ_k is the eigenvalue associated with mode *k*, k_B is the Boltzmann constant and T is temperature, assumed to be 298 K.

Detailed description about the model and the analysis procedure will appear in a manuscript in preparation by Kim, D.-N., Dietz H., and Bathe, M.

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Supplementary Methods

Thermal stability of DNA origami structures

Melting profiles for the three test structures as well as a 6 helix bundle and a 20 nucleotide DNA duplex were taken using a real time PCR machine (Stratagene MX3005P, Agilent Inc). DNA origami objects were purified by agarose gel electrophoresis and subsequent crunch and squeeze purification resulting in a yield of approximately 4 nM (~20% yield). These structures were diluted to a final concentration of ~1 nM in a total volume of 50 µl in a solution containing 16 mM $MgCl^2$, 5 mM TRIS, 1 mM EDTA, and 5 mM NaCl, as well as 1 µM of a specific intercalating dye SYBR green (Invitrogen) whose fluorescence is ~1000 times enhanced when bound to double-stranded DNA. The buffer conditions were chosen to be similar to the optimal folding conditions for all structures. The mixture was heated from 25 to 80 ^{o}C at a speed of 0.29 ^{o}C/min (1 ^{o}C every 3.5 minutes) while simultaneously recording SYBR green fluorescence. The control 20-nucleotide duplex was subjected to an identical thermal ramp at a concentration of 2.4 uM at similar buffer conditions.

Purified 18-, 24-, and 32-helix bundles in TE buffer (1mM EDTA, 10 mM TRIS) with 11 mM $MgCl^2$ were also subjected to elevated temperatures for extended periods of time using a tetrad thermal cycling engine (Biorad, formerly MJ Research). 35 µl of gel purified structures were subjected to 37, 55, and 65 ^{o}C for two hours. Structure deterioration was subsequently quantified by agarose gel electrophoresis and imaging by negative stain TEM.

Stability of multi-layer DNA origami objects against nucleases

Purified 18-, 24-, and 32-helix bundles were subjected to enzyme digestion for a series of endo- and exonucleases, specifically DNase I, T7 endonuclease I, T7 exonuclease, exonuclease I, lambda exonuclease, and Mse I. All enzymes were obtained from New England Biolabs, Ipswich, USA. Purified structures were mixed to a final concentration of ~2nM in NEB buffer #4 (New England Biolabs) containing 10U of enzyme and 8.25 mM MgCl2 (residual salt from the gel purification) in a 40 µl volume. The mixture was then incubated at 37 ^{o}C for 1 hour. Structure degradation was evaluated by both agarose gel electrophoresis and imaging by negative stain TEM. Lambda phage DNA-BstII digest was used as a control to verify enzyme activity at a concentration of 7.5 µg/ml at similar enzyme concentrations and buffer conditions. Lambda phage DNA degradation was verified by agarose gel electrophoresis.

Enzyme kinetics were measured for digestion by DNase I for both origami structures and a control duplex DNA plasmid pET24b. The origami structures were mixed at similar concentrations and buffer conditions as described above, but using 1U instead of 10 U of DNase I in order to slow down enzyme digestion.

Each reaction was set up to a total volume of 20 µl after enzyme addition and incubated at 37 °C. The enzyme solution was added sequentially in specific time intervals (see figure 5G) to each sample. All reactions were then evaluated by agarose gel electrophoresis. About 65 ng of the control plasmid pET24b was subjected to 1 U DNase I in the same, time-resolved fashion and the result was evaluated by agarose gel electrophoresis.

Stability of DNA origami structures at different buffer conditions

Structure stability was tested in the presence of 0.5X Dulbecco's Modified Eagle Medium (DMEM, common cell culture medium) by mixing 20 µl of purified structure (structures are always in a solution of TE buffer with 11 mM $MgCl^2$ directly after purification) with 20 ul of DMEM. The sample was then incubated overnight and structure degradation was evaluated by agarose gel electrophoresis. Similarly stability of test structures against crowding agents was tested by mixing 20 ul of purified structure with 20 µl of 100 mg/ml bovine serum albumin, and separately mixing 20 µl of purified structure with 20 µl 100 mg/ml dextrane. Both samples were incubated at 23 °C and samples were then imaged by TEM.

Structure stability against extremely acidic conditions (pH 2) was tested for all three structures. Since the pH could not directly be measured in the necessary small samples, a large bath (50 mL) of the post-purification buffer (pH8.3) was prepared and titrated down to pH 2 using 2.4 ml of 1M HCl. The bath was then titrated back up to pH 8 using 2.4 ml of 1M NaOH. The required volumes of acid and base were scaled down to the volume size of purified test structures. 0.96 µl of 1M HCl was added to 20 µl of purified test structures and then incubated at 37 °C for 1h. 0.96 µl of 1M NaOH was then added and samples were then evaluated by TEM.

Structures were also subjected to high salt conditions. Structure stability in 1M $MgCl^2$ conditions was tested by mixing 20 µl of purified test structure to 10 µl 3M $MgCl^2$. 1M NaCl conditions were tested by mixing 5 µl of 5M NaCl to 20 µl of purified structure. Both solutions were incubated overnight and then evaluated by TEM imaging. 50 µl of purified structures were dialyzed with ddH_20 using a dialysis membrane with 12-14 kDa cutoff. The structure was pipetted into the lid of an eppendorf tube and the membrane was then places between the lid and the tube and the lid was closed. The dialysis was performed for 6 hours with 500 ml of ddH_20, followed by exchanging the dialysis volume and dialysis for another 6 hours. 40 µl of dialyzed solution were retrieved and incubated over night at room temperature. The structures were then evaluated by direct TEM imaging.